EVAtAR-IDEA

Interactive Distributed Evolutionary Algorithm for Scripted Avatar Video Generation

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Detailed Presentation Speech Script

With Component Notes, Worked Examples, and Committee Q&A

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Domain: Interactive Distributed Evolutionary Computation (IDEC)
Target: Three peer-reviewed papers

Total delivery time:	≈ 8 minutes 15 seconds
Slides:	6 intro + interactive playground
Platforms:	HeyGen ů D-ID ů Akool
Ground truth:	AffectNet (Mollahosseini et al., 2019)
Algorithm:	Dozier (2005) IDEA

For internal use only — PhD Research Proposal Defence

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1 Slide 1 — Project Title

Delivery cue: Stand. Pause 2 seconds. Make eye contact with each committee member before you begin. This is your only chance for a first impression. Speak at 70% of your normal speed. Estimated time: 45 seconds.

Good morning, members of the committee. My name is Jahidul Arafat — I am a PhD candidate in the Department of Computer Science and Software Engineering here at Auburn University, and I am advised by Professor Gerry Dozier.

The research I am presenting today is called EVAtAR-IDEA: the Interactive Distributed Evolutionary Algorithm for Optimising Expression Tag Sequences in Scripted Avatar Video Generation.

This work sits at the intersection of three fields: evolutionary computation, affective computing, and AI-driven media production. The central question is: can a systematic algorithm automatically decide which expression tags to place at which words in a script, for how long, in order to maximise the measurable perceptual quality of an AI avatar's facial expression?

The research targets three peer-reviewed papers and tests the algorithm across three avatar platforms — HeyGen, D-ID, and Akool. All scoring uses AffectNet as ground truth: the largest publicly available facial expression database in the wild, published in IEEE Transactions on Affective Computing in 2019, with over one million images and 450,000 manually annotated faces across eight emotion classes.

Component Notes

What is AffectNet?

Mollahosseini, Hasani, & Mahoor (2019). IEEE TAFFC 10(1), 18–31. DOI: 10.1109/TAFFC.2017.2740923
The eight categorical classes (Table 3 of the paper): Neutral, Happy, Sad, Surprise, Fear, Anger, Disgust, Contempt. Note: AffectNet also defines "None", "Uncertain", and "Non-Face" categories, but only the eight expression classes are used for classification. DeepFace (Serengil & Ozpinar, 2020) uses AffectNet weights as its emotion recognition backbone.

What is IDEA?

Interactive Distributed Evolutionary Algorithm. Introduced by Dozier (2005) in the IDEC (Interactive Distributed Evolutionary Computation) domain. EVAtAR-IDEA is the first application of IDEA to affective avatar expression optimisation.

2 Slide 2 — The Problem

Delivery cue: Slow down here. The committee needs to feel the problem before hearing the solution. Speak in first person throughout. Pause 1 second after every statistic. Let the numbers land. Estimated time: 90 seconds.

Let me ground this research in a concrete, first-hand experience.

During February to May 2026, I was producing scripted speeches using Aubria — HeyGen's AI avatar — as part of the AI@AU/Team Since Series initiative at Auburn University. Aubria delivered keynote speeches at Auburn Board of Trustees meetings, programme briefings for the DCSITE detection canine science initiative, and the AI Ethics Iron Bowl event, where Aubria addressed students, faculty, and industry professionals live. A typical speech script is 500 to 800 words long. Each word is a potential tag placement point. Tags like [serious], [excited], [pause: 2s], placed manually, word by word, clip by clip.

The problem I encountered was that every attempt broke something different. In one iteration, the avatar's face would freeze mid-sentence — a classic F1 "no emotion tag" failure the "no emotion tag" failure, where no expression tag was present at all, where no emotion tag was present for the first half of the clip. In another, the expression would peak too late — I placed [excited] at word 14 of a 16-word sentence, but the avatar's rendering engine needs to fire the expression before the corresponding audio peak. That is the F3 the late-onset failure mode, where the expression fires after the audio peak.

In the worst cases, I placed two emotion tags within three words of each other. For example: [serious] at word 4 and [excited] at word 6. The avatar engine cannot transition between two emotion states that close together — both degrade into a neutral blank expression. That is F2 tag stacking, where two emotion tags are placed too close together — one of eight failure modes I later formalised.

The result, in aggregate: more than 200 hours of manual iteration, across more than 100 re-renders, to produce a single polished video. Token cost: approximately 12 HeyGen credits per clip, at $0.004 per credit = $0.048 per render. For a 500-word script, that is 35 clips. At 100 re-renders per clip, the token cost exceeds $166 per video.

Now, HeyGen offers Auto-Enhance: a button that places tags automatically. I want to be precise about why that is insufficient. Auto-Enhance places tags typically at 30%, 65%, and 80% of sentence length. The 65% and 80% positions almost always trigger F3 late-onset late-onset meaning the expression fires after the audio peak

meaning the expression fires after the audio has already peaked — the expression fires after the audio peak. There is no perceptual fitness function, no AffectNet validation, no failure taxonomy, and no published methodology.

Worked example: Given the sentence "In twelve months, three papers, one hundred and sixty-six dollars, two hundred hours — and it is finally done," Auto-Enhance places [enthusiastic] at word 3 and again at word 13 — position 68% of 19 words. F3 late-onset meaning the expression fires after the audio peak meaning the expression fires after the audio has already peaked is triggered because the emotional peak arrives at word 16: "finally done." The avatar expresses enthusiasm before the sentence reaches its emotional content.

EVAtAR-IDEA addresses this systematically — not with rules, but with rigorous evolutionary search, a pre-registered fitness function, and AffectNet validation.

The Eight Failure Modes (F1–F8 the sub-perceptual failure)

Code	Name	Criterion / Example
F1	No emotion tag	No T1 token present; avatar renders neutral Example: "Today we will explore calculus." — no tag
F2	Tag stacking	Two emotion tags within 2 word positions Example: [serious] word 4 + [excited] word 6 both degrade
F3	Late onset	Primary tag fires after 60% of sentence length Example: 12-word sentence, tag at word 9 = 75% F3 triggered
F4	Overloading	More than 3 emotion tags in one clip
F5	Gesture overload	More than 2 gesture tokens in one clip
F6	Amplitude fatigue	[excited] used more than twice in one clip
F7	Timing overflow	Total pause duration exceeds 3 seconds
F8	Sub-perceptual	[curious] Fear proxy; mean f1 = 0.31 Note: [curious] is not an AffectNet category — researcher inference

Critical transparency note for committee (F8 the sub-perceptual failure): The HeyGen tag [curious] has no equivalent in AffectNet's 8-class taxonomy (Table 3, Mollahosseini et al. 2019). EVAtAR-IDEA maps it to the Fear class as the closest proxy in Russell's circumplex: curiosity shares high arousal and low-to-negative dominance with Fear. This is a researcher inference, not an AffectNet definition. It is pre-registered as a threat to validity in Paper 1.

Auto-Enhance vs. EVAtAR-IDEA

HeyGen Auto-Enhance	EVAtAR-IDEA
Rule-based / LLM-prompted	Evolutionary search; 21-token vocabulary
No perceptual fitness function	$F(p) = 0.5 \cdot f_1 + 0.3 \cdot f_2 + 0.2 \cdot f_3$
Not AffectNet validated	DeepFace + AffectNet 8-class scoring
No failure taxonomy applied	Constraint repair for F2, F3, F5
No pre-registration	Wilcoxon + Cohen d + OSF pre-registration
Heuristic tag placement (30/65/80%)	Optimised position from 1,000+ search space
HeyGen-only	P3: HeyGen $\star\star\star$, D-ID $\star\star$, Akool $\star$

3 Slide 3 — Research Objective

Delivery cue: Technical but accessible. Write the formula on the board if available. Emphasise "pre-registered" as a methodological strength — it protects both you and the finding. Estimated time: 75 seconds.

The core insight of EVAtAR-IDEA is that manual expression tag placement is, at its heart, a combinatorial optimisation problem.

Given a sentence, a vocabulary of 21 confirmed expression tokens, the possible word positions, and hold durations from 0.5 to 3 seconds, the search space exceeds one thousand distinct combinations per sentence. No human can explore that space systematically.

I formalise the optimisation target as:

$$F(p) = 0.5 \cdot f_1 + 0.3 \cdot f_2 + 0.2 \cdot f_3$$

Where:

- f_1 = AffectNet class accuracy: mean probability across all video frames that DeepFace detects the intended emotion class. Example: for a sentence tagged [excited], f_1 = mean P(Happy) from DeepFace, because [excited] maps to AffectNet Happy.
- f_2 = Timing alignment: does the peak of the detected expression coincide with the audio peak position in the sentence?
- f_3 = Expression consistency: is the detected emotion stable across all frames, or does it flicker below the threshold mid-sentence?

The weights — 50%, 30%, 20% — are pre-registered at OSF before data collection. Getting the right emotion class is the primary criterion. Timing is secondary. Consistency is tertiary. Paper 2 includes a sensitivity analysis.

Three baselines are compared: p_{auto} (Auto-Enhance), p_{manual} (expert human), and p^* (EVAtAR-IDEA). Pre-registered success threshold: $F(p^*) \geq 0.75$, Wilcoxon signed-rank $p < 0.05$, Cohen's $d \geq 0.5$.

The 21-Token Vocabulary

The vocabulary was derived empirically before P1 data collection:

Tier	Count	Tokens & Notes
T1 AffectNet-Confirmed	6	[excited]→Happy, [serious]→Neutral, [surprised]→Surprise, [curious]→Fear*, [sad]→Sad, [angry]→Anger
T2 Excluded	6	[warm], [confident], [thoughtful], [calm], [nervous], [frustrated] — each excluded with documented reason
T3 Gesture/Delivery/Timing	9+	[smile], [nod], [lean:forward], [eyebrow:raise], [head:tilt], [emphasis], [pause:0.5s–3s], etc.

*[curious] mapped to Fear by researcher inference; not an AffectNet definition.

4 Slide 4 — Three Prototypes ů Three Papers

Delivery cue: Pause between each prototype. Point to P1, P2, P3 on screen. Emphasise "sequential" — P2 cannot begin without P1 data; P3 cannot begin without P2 sequences. Estimated time: 105 seconds.

EVAtAR-IDEA is structured across three sequential prototypes, each targeting a peer-reviewed paper.

Prototype 1 — Baseline Measurement. I collect 50 sentences across the 8 AffectNet emotion classes. Each sentence is assigned a difficulty level using a three-factor rubric: (1) Flesch-Kincaid grade for syntactic complexity; (2) Russell & Mehrabian's VAD model for emotional ambiguity; (3) F3 late-onset meaning the expression fires after the audio peak meaning the expression fires after the audio has already peaked risk from the position of the primary emotion cue word.

Worked example: "Today we will explore the fundamental theorem of calculus" — 9 words, FK grade 8, emotion cue at word 4 = 44% of sentence, low F3 late-onset late-onset late-onset meaning the expression fires after the audio peak risk how likely the expression is to fire after the audio peak ⇒ Easy. Versus: "In twelve months, three papers, two hundred hours — and it is finally done" — 19 words, FK grade 11, emotion cue at word 16 = 84% ⇒ Hard.

Transparency note: [curious] has no AffectNet equivalent. Mapped to Fear as closest VAD proxy. Pre-registered as threat to validity. Output: D1 dataset. Targets Paper 1: Vocabulary and Corpus Study.

Prototype 2 — EVAtAR-IDEA Engine. For each of 50 sentences: initialise 8 random candidate sequences, evolve across 8 generations. Each generation: render 8 clips via HeyGen (≈60s per render), score via DeepFace (AffectNet weights), tournament select ($k = 3$), mutate (shift_pos, change_tag, add, remove, change_dur), constraint repair (fix F2 tag-stacking tag-stacking and F3 late-onset meaning the expression fires after the audio peak meaning the expression fires after the audio has already peaked violations). That is 64 renders per sentence. Sequences with $F(p^*) \geq 0.70$ enter the Meme Space, seeding same-emotion sentences and reducing cost from ∼64 to ∼20 renders. Targets Paper 2: Optimisation Study.

Prototype 3 — Cross-Platform Portability. The same p^* from P2 is tested on HeyGen (inline tags, full control), D-ID (NL motion hints, expected 15–25% degradation), and Akool (audio-driven only, lower bound). Same DeepFace scoring pipeline throughout. Answers: is EVAtAR-IDEA platform-specific or generalisable? Targets Paper 3: Portability Study.

GA Parameter Table

Parameter	Value	Justification
Population size	8	Balances diversity with render cost (64 renders/gen)
Generations	8	Sufficient convergence observed in pilot; pre-registered
Tournament k	3	Maintains diversity while applying selection pressure
Meme threshold	$F(p^*) \geq 0.70$	Conservative; below pre-registered success threshold
Mutation ops	5	shift_pos, change_tag, add_tag, remove_tag, change_dur
Constraint repair	Post-mutation	Fixes F2 stacking and F3 late-onset violations

Corpus Difficulty Rubric (Paper 1)

Level	Word count	FK grade	F3 risk / cue position
Easy	≤ 10	≤ 6	Cue $< 40\%$ of sentence
Medium	11–16	7–10	Cue 40–65%
Hard	≥ 17	≥ 11	Cue $> 65\%$ or multiple competing cues

References: Flesch (1948); Russell & Mehrabian (1977); Mohammad & Turney (2013) NRC Lexicon.

5 Slide 5 — What You Are About to Experience

Delivery cue: Conversational. You are a guide, not a lecturer. Invite the committee to follow along. Make it feel like an experience. Estimated time: 90 seconds.

Before you interact with the playground, let me map what each stage demonstrates and which paper it corresponds to.

Stage 1 — Select Script (P1 corpus collection): Choose from 32 sentences, 4 per emotion category, across all 8 AffectNet classes. When you select a category, three panels appear: the corpus design rationale explaining why 8 categories and how difficulty was assigned; a paragraph-to-clips explainer showing how a real Auburn Canine Science script breaks into 35 individual render units; and a live Auto-Enhance preview showing exactly which tags HeyGen would place and which failure modes they would trigger.

For example, selecting "Rhetorical" and picking the sentence "Can a computer really learn to be more expressive than a human researcher?" — Auto-Enhance would place [curious] at word 3 and [thoughtful] at word 9. The preview flags this as F2 near-stacking, where two tags are placed dangerously close together two tags placed dangerously close together and F3 moderate risk, with a simulated F(p_{auto}) of 0.57.

Stage 2 — Build Tags Manually (P1 baseline): Click any word to add a tag. The F1 through F8, our eight documented avatar expression failure modes the sub-perceptual failure where a tag produces too weak a signal for DeepFace to detect reliably eight documented failure modes failure taxonomy panel on the right updates in real time. Place a tag at position 9 of a 12-word sentence and F3 late-onset meaning the expression fires after the audio peak meaning the expression fires after the audio has already peaked fires immediately — you watch it happen. Your final $F(p_{manual})$ is stored as the baseline the GA must beat.

Stage 3 — Run GA (P2): Choose 1–5 independent runs. Watch convergence across 8 generations. The red dashed line is your $F(p_{manual})$. The Meme Space fills as solutions exceed 0.70. Each generation shows the mutation applied and the best candidate sequence.

Stage 4 — Results (P2 + P3): Three-way comparison table: Auto-Enhance vs your manual sequence vs GA-optimised p^*. Pre-registered threshold check. Cross-platform projection for HeyGen, D-ID, and Akool. D1 CSV export.

6 Slide 6 — Full Pipeline, Pseudocode, and Algorithm

Delivery cue: Use the three tab buttons — Pipeline SVG (30s), Pseudocode (30s), Plain English (30s). Never read pseudocode line by line; summarise the structure. Estimated time: 90 seconds.

This slide presents the complete EVAtAR-IDEA algorithm from three angles.

Pipeline View — left to right. P1, in orange on the left, takes the corpus, runs both Auto-Enhance and human labelling, renders both via HeyGen, scores both via DeepFace, and produces the D1 dataset. The baseline arrow carries $F(p_{manual})$ and $F(p_{auto})$ into P2. P2, in the central green box, contains the inner GA loop inside a dashed border — 64 renders per sentence. Each of 8 candidates is rendered, scored, selected, mutated, and repaired. Sequences above 0.70 enter the Meme Space. After 8 generations, p^* moves right into P3, where the same sequence is tested on HeyGen, D-ID, and Akool. The pre-registered threshold dashed line at the bottom — $F(p^*) \geq 0.75$, Wilcoxon $p < 0.05$, Cohen's $d \geq 0.5$ — is the primary success criterion for Paper 2.

Pseudocode View. The algorithm opens with PRE-REGISTER at OSF before any data collection. P1 runs AutoEnhance and Human.LabelTags in parallel, renders and scores both. P2's inner loop: for each candidate, render, score, store if ≥ 0.70, tournament select, mutate, constraint repair. The final line is a three-way Wilcoxon signed-rank test. P3 applies p^* to three platforms and reports degradation.

Plain English View. P1 asks: where does human expertise break down, and where does Auto-Enhance fail? P2 asks: can an evolutionary algorithm beat both, by a statistically meaningful margin? P3 asks: does that improvement transfer across platforms, or is it locked to HeyGen?

This is the first application of Dozier's 2005 IDEA framework to affective avatar expression optimisation — and the first work to formally compare evolutionary tag optimisation against a commercially deployed heuristic baseline.

7 Anticipated Committee Questions and Responses — Complete

Section A — Research Design & Methodology

Q1 — Why exactly 8 emotion categories? AffectNet defines exactly 8 classes in its categorical model: Neutral, Happy, Sad, Surprise, Fear, Anger, Disgust, Contempt (Table 3, Mollahosseini et al., 2019). DeepFace uses these exact 8 classes. A 9th category would have no f_1 score — the scoring oracle would break. The categories are determined by the measurement infrastructure, not by convenience.

Q2 — Why F(p) weights 0.5, 0.3, 0.2 and not equal thirds? Pre-registered before data collection. f_1 at 50%: wrong emotion class is categorical failure regardless of timing. f_2 at 30%: timing determines perceived synchrony. f_3 at 20%: flickering is visible but less disruptive. Paper 2 includes sensitivity analysis varying each weight by ± 0.1.

Q3 — Why not a data-driven learned reward model instead of hand-designed F(p)? A learned reward model requires a large labelled dataset of human perceptual ratings of avatar expressions — that dataset does not exist yet. The hand-designed formula with pre-registered weights is methodologically stronger: it makes the success criterion falsifiable before data collection, prevents post-hoc weight tuning, and produces an interpretable decomposition. A learned model would be a black box the committee cannot interrogate.

Q4 — You map [curious] to Fear — how do you defend that? [curious] is not in AffectNet. I chose Fear as the closest proxy using Russell's VAD circumplex: both share high arousal and low-to-negative dominance. This is a researcher inference, pre-registered as a threat to validity in Paper 1. The GA compensates by pairing [curious] with [head:tilt] and [eyebrow:raise] tokens.

Q5 — Is there circularity — F1–F8 failure modes derived from the same observations? The F1–F8 taxonomy was derived from exploratory production work (Feb–May 2026), then treated as a fixed pre-registered framework before P1 data collection. Critically, the taxonomy informs the constraint repair module and diagnostic reporting — it does not compute $F(p)$. The fitness function is entirely AffectNet-grounded and independent of the taxonomy. Search space and evaluation are cleanly separated.

Q6 — How do you account for inter-rater reliability in difficulty assignments? The difficulty rubric is algorithmic, not subjective — it applies the Flesch-Kincaid formula, NRC Emotion Lexicon VAD scores, and a deterministic cue-position calculation. These

are reproducible computations. Any researcher applying the same rubric to the same sentence obtains the same difficulty label. No inter-rater variance arises.

Section B — Algorithm Design

Q7 — Why IDEA (Dozier 2005) and not NSGA-II or CMA-ES? Three reasons: (1) IDEA is designed for Interactive Distributed Evolutionary Computation, explicitly incorporating human-in-the-loop interaction — architecturally consistent with EVAtAR-IDEA's design. (2) Prof. Dozier, my supervisor, authored IDEA. (3) The tag sequence problem has a small, structured search space that does not require CMA-ES parameter adaptation or NSGA-II multi-objective decomposition.

Q8 — Why tournament selection k=3? With population size 8, $k = 3$ samples 37.5% per tournament — sufficient selection pressure without eliminating diversity. Roulette wheel selection is near-uniform for fitness values in the 0.50–0.85 range. The constraint repair ensures weak candidates are corrected before competition, maintaining diversity.

Q9 — Why 8 generations? Determined by two constraints: (1) render cost — 8 gens $\times$ 8 pop = 64 renders per sentence $\times$ \$0.048 = \$3.07/sentence, acceptable for a 50-sentence corpus; (2) pilot testing showed convergence by generation 5–6 for this vocabulary size. Paper 2 reports convergence curves.

Q10 — Mutation rate and convergence sensitivity? Each generation applies exactly one mutation operator (uniformly selected from 5) per survivor — effectively mutation probability = 1.0. Standard for small populations requiring diversity maintenance. Sensitivity analysis in Paper 2 reports convergence curves across 5 trial runs. Constraint repair acts as a second-order selection mechanism — invalid mutations are repaired rather than discarded.

Q11 — Meme Space threshold 0.70 vs success threshold 0.75 — why the gap? The 0.05 gap is deliberate. If the Meme Space only stored solutions above 0.75, it would rarely fill during early runs. The conservative 0.70 threshold ensures stored solutions are genuinely good (well above 0.50 neutral baseline) while giving the GA room to improve seeded solutions toward 0.75 through further mutation.

Q12 — What if the GA finds p* but F(p*) < F(p_manual)? This is a valid and reportable outcome — documented in D2 as a negative case. The Wilcoxon test operates on the full distribution across 50 sentences; a few negative cases contribute to variance estimation, not invalidation. Negative cases clustered by emotion class (e.g., [curious] sentences) would themselves be informative findings reported in the per-class failure analysis of Paper 2.

Section C — Evaluation & Statistics

Q13 — What is the statistical power of n=50? G*Power 3.1 Wilcoxon signed-rank: $n = 50$, two-tailed $\alpha = 0.05$, $d = 0.5$ (pre-registered medium effect) yields power ≈ 0.78 — below the 0.80 convention. This limitation is acknowledged and pre-registered. Mitigations: (1) $d = 0.5$ is conservative; pilot observations suggest larger effect; (2) corpus is stratified across 8 classes and 3 difficulty levels for variance coverage; (3) class-level analysis increases local power to $n \approx 6$ per emotion class.

Q14 — Why Wilcoxon signed-rank rather than a paired t-test? $F(p)$ scores are bounded in $[0, 1]$ and unlikely to be normally distributed — particularly for extreme failure cases where $F(p_{manual})$ clusters near 0.3–0.5. Wilcoxon makes no distributional assumptions and is standard for paired bounded continuous data in HCI and affective computing evaluation. A paired t-test is included as a secondary analysis.

Q15 — How do you validate DeepFace/AffectNet scores against human perception? AffectNet was validated against 60.7% inter-annotator agreement across 36,000 images (Table 6, Mollahosseini et al. 2019). DeepFace achieves 0.64 normalised F1-score on this test set (Table 9). I use $F(p)$ as a proxy for perceptual quality, not a direct measure. Paper 1 includes a perceptual validation: 10 human raters view a subset of clips and rate expression quality on a Likert scale. Spearman correlation between $F(p)$ and human ratings is reported.

Q16 — Is the Auto-Enhance comparison fair? Yes, for the following reasons: Auto-Enhance is the tool practitioners actually use when they choose not to tag manually — it is the de facto alternative baseline. The comparison is not "EVAtAR-IDEA beats a research system" but "EVAtAR-IDEA outperforms the best available automated alternative." Auto-Enhance's behaviour is documented empirically. Its algorithm being unpublished means the comparison is behavioural, not architectural.

Section D — Technical & Scope

Q17 — What if HeyGen changes their API? Mitigations: (1) all P1/P2 data collection is scheduled for a defined time window (2026) to minimise API version variance; (2) API version is documented as a methodological variable; (3) $F(p)$ is platform-agnostic; (4) P3 explicitly tests D-ID and Akool, demonstrating framework independence from HeyGen.

Q18 — How does the research scale to other avatars? Scalability is P3's contribution. If $F(p)$ degrades gracefully ($\leq$25%) across D-ID and Akool, the framework generalises. Beyond P3: any scriptable expression interface (Synthesia, Colossyan, Unity avatars) could

adopt the same vocabulary, fitness function, and evolutionary architecture. The 21-token vocabulary would require re-validation per platform.

Q19 — How do you define word position? Token index in the whitespace-tokenised sentence. Each whitespace-separated token = one position, regardless of syllable count. Contractions treated as single tokens. Punctuation stripped before tokenisation. Consistent with HeyGen's script parser. Documented as a methodological simplification.

Q20 — Does tag order matter? Can two tags share the same position? Two tags at the same position are valid in the representation (list of position, tag, duration tuples — not a dictionary). Two emotion tags at the same position trigger F2 tag-stacking and are repaired by constraint repair. An emotion tag + gesture tag at the same position (e.g., [excited] + [smile] at position 2) is valid and often beneficial for f_1.

Q21 — What is the novelty claim vs existing work? Three novel claims: (1) First application of Dozier (2005) IDEA to avatar expression optimisation in the IDEC domain. (2) First systematic F1–F8 failure taxonomy for scripted avatar expression tags, grounded in empirical production experience and AffectNet-validated. (3) First three-way empirical comparison of evolutionary optimisation vs human expert vs commercial heuristic (Auto-Enhance) using a pre-registered AffectNet-grounded fitness function. Existing avatar expression control work (gesture synthesis, head motion) does not address the expression tag placement problem for commercial text-to-avatar platforms.

Section E — Pre-Registration & OSF

Q22 — Are P1 and P2 pre-registrations separate? Yes. P1 pre-reg: filed before any data collection; covers 8 categories, 50-sentence corpus, difficulty rubric, F1–F8 taxonomy, $F(p)$ formula and weights, [curious]→Fear mapping as threat to validity. P2 pre-reg: filed after P1 data collection is complete; covers GA parameters (pop=8, gen=8, $k = 3$), Meme Space threshold (0.70), success criterion ($F(p^*) \geq 0.75$), Wilcoxon+Cohen d thresholds, three-way comparison design. Separating them prevents P1 data from being used to optimise P2 parameters post-hoc.

Q23 — What is pre-registered vs decided after data collection? Pre-registered before P1: measurement framework. Pre-registered before P2: all GA parameters and statistical thresholds. Decided after P1 (documented as exploratory): secondary analyses, per-class breakdowns, meme space efficiency observations. This separation is standard and explicitly stated in both OSF filings.

Section F — Playground Questions

Q24 — Why are playground scores simulated? The playground is explicitly labelled throughout as a simulation with research-validated mock values. Live HeyGen calls would: (1) expose API authentication tokens; (2) take 30–90 seconds per render, making the playground unusable for committee demonstration. Mock values are calibrated from observed DeepFace output distributions. The architectural logic is identical to the real system.

Q25 — Why 32 sentences in playground but 50 in real study? 32 sentences (4 per category $\times$ 8 categories) is a completable representative subset for a committee session ($\sim$5–10 minutes per sentence to tag, render, assess, and run GA). The full 50-sentence corpus includes additional sentences per class for statistical power and edge-case coverage.

Q26 — Is the GA exploiting the mock data? No. Mock $F(p)$ values include calibrated stochastic noise matching real DeepFace variance per emotion class and difficulty level. The GA calls the same fitness computation as the real system. Crucially, the mock data is calibrated to show realistic improvement curves — for Hard sentences in the [curious] category (F8 risk), the playground accurately shows the GA struggling to exceed 0.68–0.72, reflecting the real research challenge.

7 Key References

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